



Multinomial Logistic Regression



Multinomial Logistic Regression

What if the response has $K > 2$ classes?

Medical condition at emergency room: stroke, overdose, or seizure?

Multinomial Logistic Regression is a straightforward generalization of logistic regression for $K > 2$



Generalization of Logistic Regression

We choose the k -th class as the *baseline*, then for $k = 1, \dots, K - 1$:

$$Pr(Y = k|X = x) = \frac{e^{\beta_{k0} + \beta_{k1}x_1 + \dots + \beta_{kp}x_p}}{1 + \sum_{l=1}^{K-1} e^{\beta_{l0} + \beta_{l1}x_1 + \dots + \beta_{lp}x_p}}$$

For $k = K$

$$Pr(Y = K|X = x) = \frac{1}{1 + \sum_{l=1}^{K-1} e^{\beta_{l0} + \beta_{l1}x_1 + \dots + \beta_{lp}x_p}}$$

Log odds:

$$\log \left(\frac{Pr(Y=k|X=x)}{Pr(Y=K|X=x)} \right) = \beta_{k0} + \beta_{k1}x_1 + \dots + \beta_{kp}x_p$$



Does the Choice of the Baseline Matter?

No, the coefficients change with a different baseline, but the log odds between pairs of classes stay the same

The meaning of the coefficients depends on the baseline



The *softmax* encoding

No baseline is selected, all classes are treated symmetrically:

$$Pr(Y = k|X = x) = \frac{e^{\beta_{k0} + \beta_{k1}x_1 + \dots + \beta_{kp}x_p}}{\sum_{l=1}^K e^{\beta_{l0} + \beta_{l1}x_1 + \dots + \beta_{lp}x_p}}$$

$$\log \left(\frac{Pr(Y=k|X=x)}{Pr(Y=k'|X=x)} \right) = (\beta_{k0} - \beta_{k'0}) + (\beta_{k1}x_1 - \beta_{k'1}) + \dots + (\beta_{kp}x_p - \beta_{k'p})$$



Generative Models for Classification



This is not about LLMs, sorry



Problems with Logistic Regression?

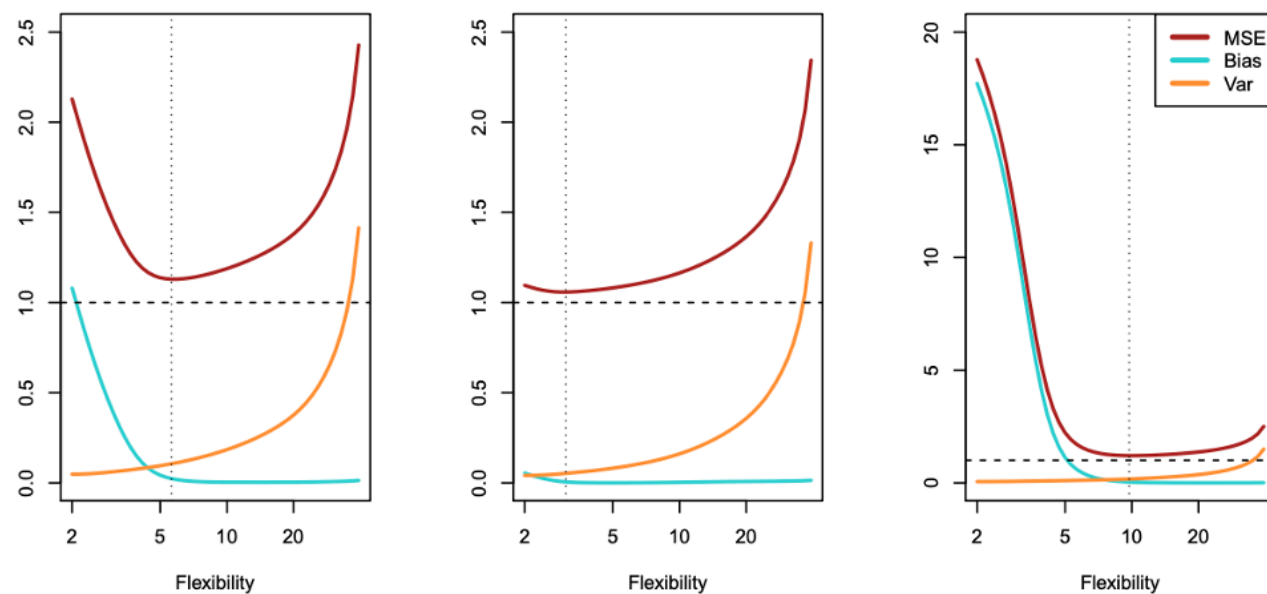
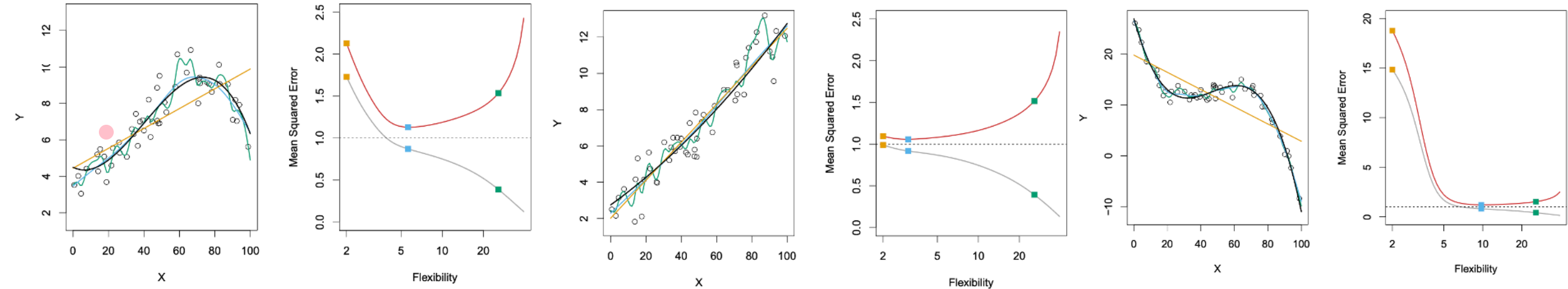
Logistic regression models directly the conditional distribution of the response Y given the predictor(s) X

$$Pr(Y = k | X = x)$$

If the separation between the two classes is substantial, the coefficients in logistic regression are unstable

Extension to $K > 2$ classes looks a bit “artificial”

Other methods are more accurate if the distribution of the predictors is normal



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$$E \left(y_0 - \hat{f}(x_0) \right)^2 = \text{Var}(\hat{f}(x_0)) + [\text{Bias}(\hat{f}(x_0))]^2 + \text{Var}(\epsilon).$$



(Very Simple) Question

Let's assume that we know the probability distribution from which our data are generated $P(Y|X)$

Given this knowledge, what is the best classifier that we can build?



Theoretical Result: the Bayes Classifier (from 1st lecture)

The test error is minimized by a very simple classifier that assigns each observation to the most likely class, given its predictor values

That is, we should simply assign the observation with predictor x_0 to the class j for which the following probability is largest:

$$Pr(Y = j | X = x_0)$$

The error made by the Bayes classifier is only the *irreducible* error:

$$1 - E(\max_j \{Pr(Y = j | X)\})$$



Why the Bayes Classifier?

Usually, we only have observations X , but we do not know the distributions that generates them

So, the Bayes classifier remains an *unattainable* gold standard

When we simulate data from a given distribution, then we can also calculate this *gold standard*



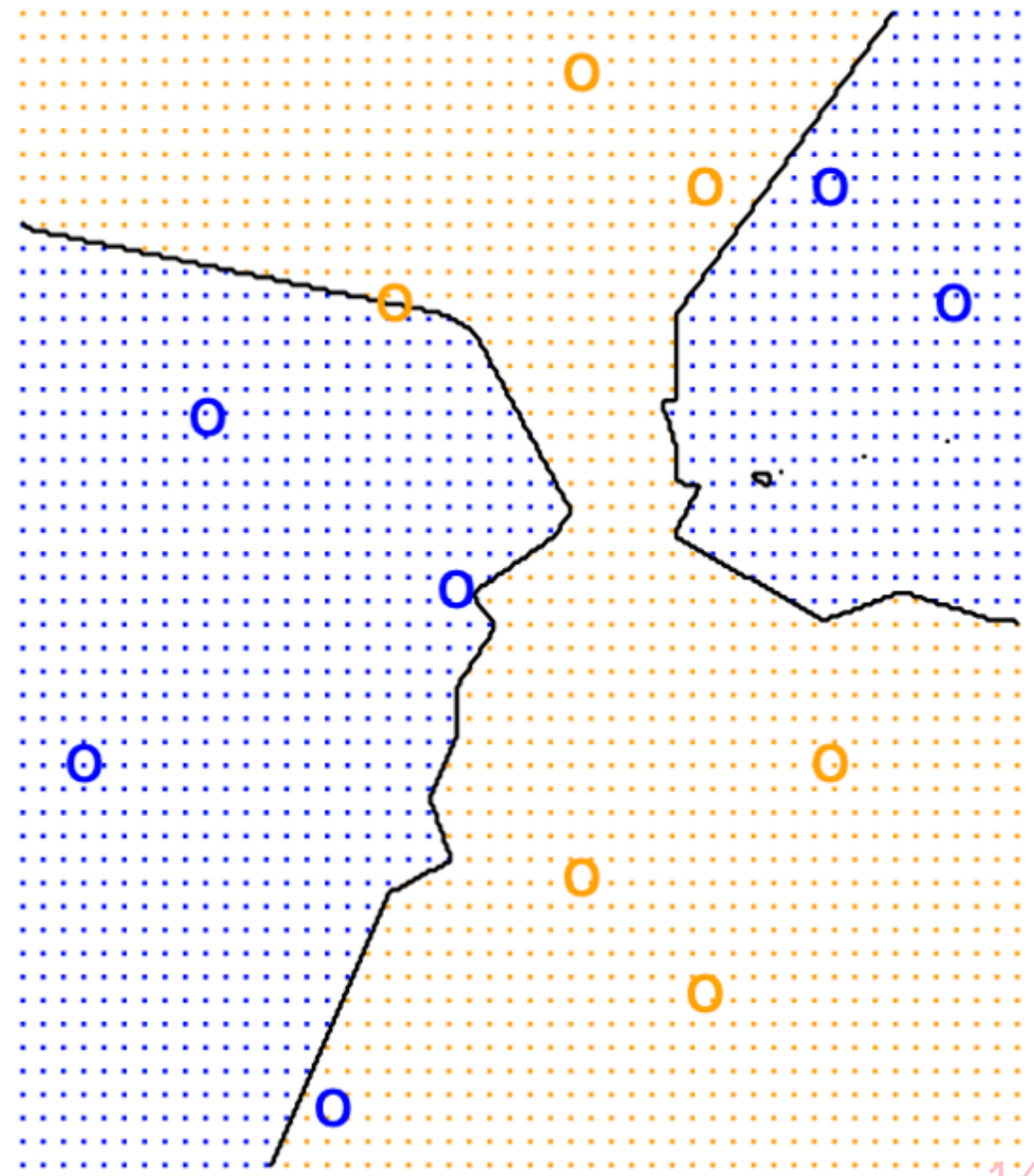
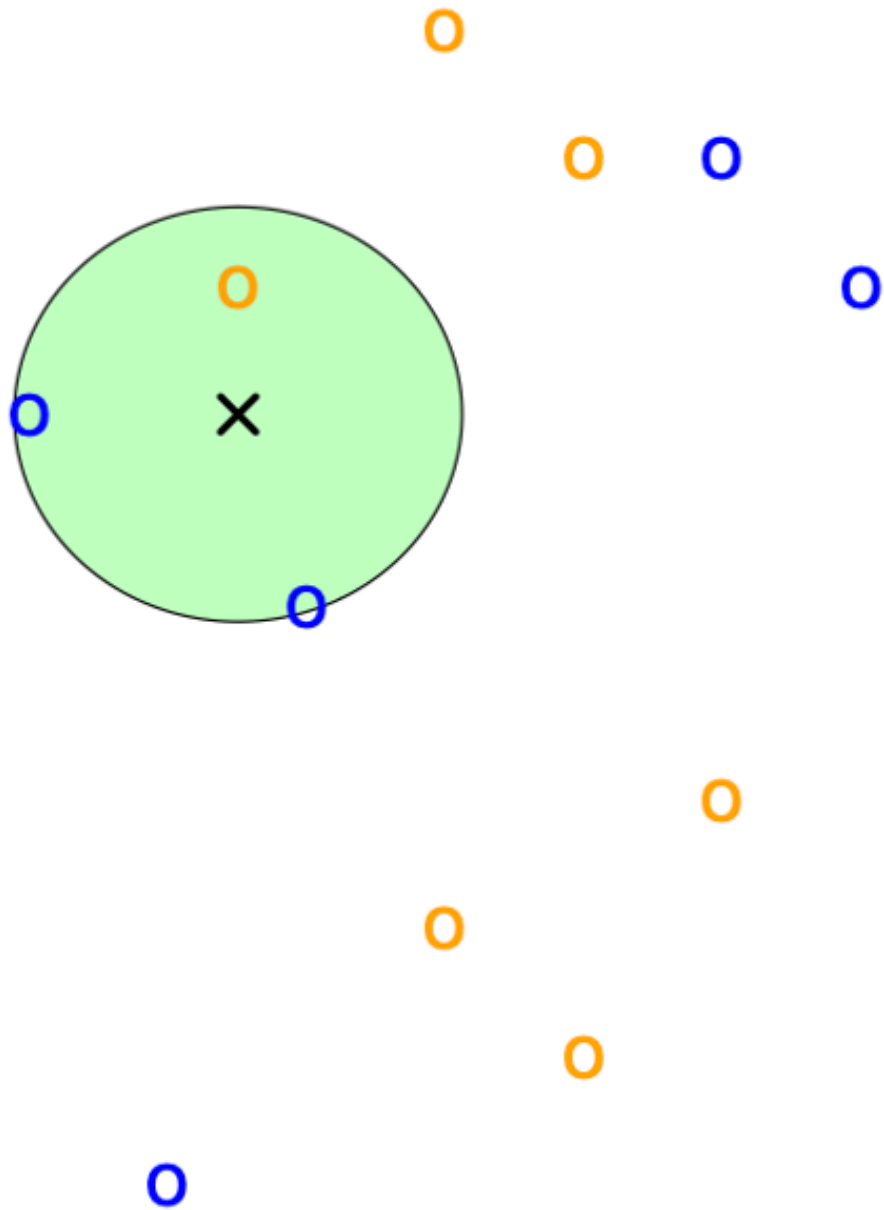
KNN Classifier (K-Nearest Neighbors)

We do not know the conditional distribution of Y given X , but we can estimate it 😊, and then use our estimate as a Bayes classifier

Given a test observation x_0 , the KNN classifier first identifies the K points ($\mathcal{N}_0$) in the training data set closest to x_0 . Then, conditional probability for class j is estimated as the fraction of points in ($\mathcal{N}_0$) with response equal to j

The class j associated with the largest conditional probability is chosen to classify x_0

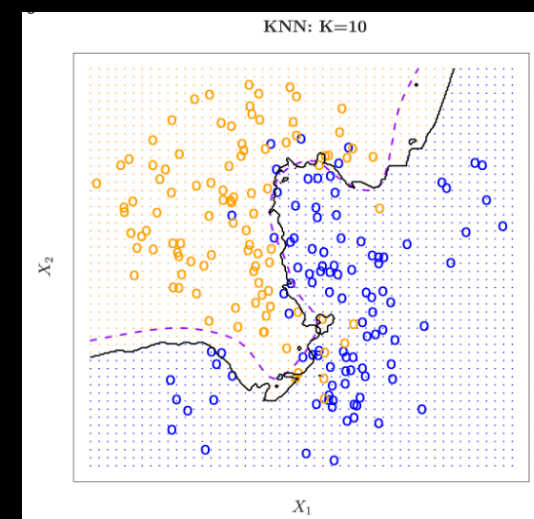
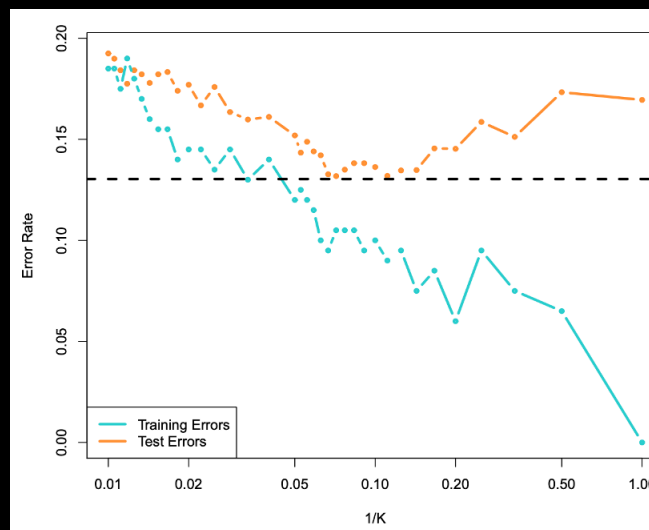
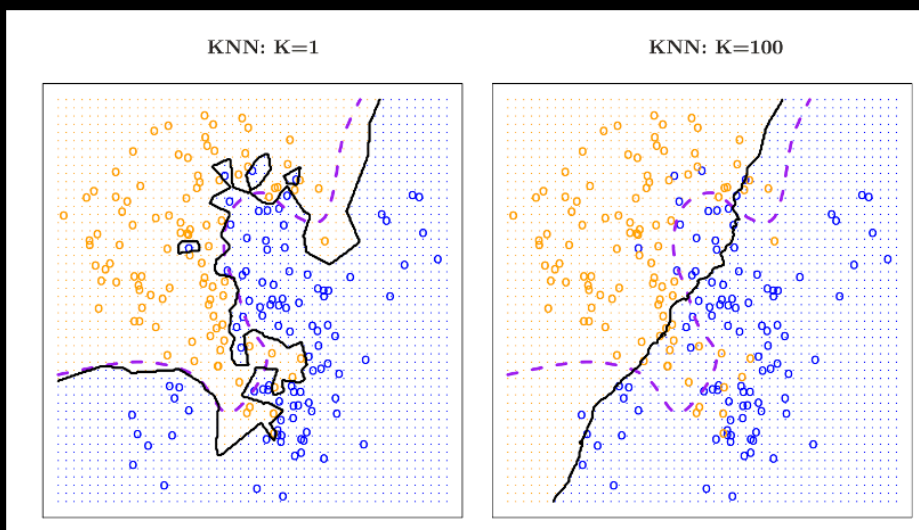
$$Pr(Y = j|X = x_0) = \frac{1}{K} \sum_{i \in \mathcal{N}_0} I(y_i = j)$$





KNN v. Bayes Decision Boundaries

Lower $K \longrightarrow$ More Flexible Model





The Intuition of Generative Models

Propose a more sophisticated way to estimate $Pr(Y = k|X = x)$ than KNN

But how?

Model the distribution of the predictors X separately for each value of Y (class)

Use the *Bayes Theorem* to flip the predictor distribution into:

$$Pr(Y = k|X = x)$$



The Bayes Theorem

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

$P(A|B)$: conditional (or “posterior”) probability of A given B

$P(B|A)$: conditional probability of B given A , or the “likelihood” of A given a fixed B

$P(A)$ and $P(B)$: chance of observing A and B , respectively (prior or marginal probabilities)



The Bayes Theorem: Example

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

A driving alcohol test gives 90% true positive results: *if you are drunk, you test positive 90% of the times*

The test also gives 80% true negatives: *if you are not drunk, you test negative 80% of the times*

20% of false positives : *you are not drunk, but test positive 20% of the times*

Assume 5% of the drivers in Ulsan are drunk

... what is the probability that driver that tests positive is actually drunk?



The Bayes Theorem: Example

A : driver is drunk, B : driver test positive

$$\text{Bayes theorem: } P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

$$P(A) = P(A|B) + P(A|\neg B)$$

$$P(\text{drunk}|\text{pos}) = \frac{P(\text{pos}|\text{drunk})P(\text{drunk})}{P(\text{pos})} = \frac{P(\text{pos}|\text{drunk})P(\text{drunk})}{P(\text{pos}|\text{drunk})P(\text{drunk}) + P(\text{pos}|\text{non-drunk})P(\text{non-drunk})}$$

$$P(\text{drunk}|\text{pos}) = \frac{0.9 \times 0.05}{0.9 \times 0.05 + 0.2 \times 0.95} = 19\%$$

Bayes Theorem helps to turn the prior probabilities (is a drunk or sober driver testing positive?) into the posterior probability (is a driver who tested positive actually drunk?)



Intuition of Generative Models

Let $Y = k$ (the observation belongs to class k) be event A

Let $X = x$ (the predictors for an observation are equal to x) be event B

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Which we can rewrite as: $Pr(Y = k|X = x) = \frac{\pi_k f_k(x)}{\sum_{l=1}^K \pi_l f_l(x)}$

π_k : overall (prior) probability that a randomly chosen observation belongs to class k

$f_k(x)$ = density function of X for an observation belonging to class k

$$f_k(x) = Pr(x|Y = k)$$



So what? Use the Right-Hand Side

$$Pr(Y = k|X = x) = \frac{\pi_k f_k(x)}{\sum_{l=1}^K \pi_l f_l(x)}$$

Logistic Regression models directly the posterior probability $Pr(Y = k|X = x)$

Generative models estimate π_k and $f_k(x)$

Estimating π_k is *easy*: compute the fraction of the sample that belongs to class k

Estimating $f_k(x)$ is **hard**!



Three Methods for Estimating $f_k(X)$

Linear Discriminant Analysis (LDA)

Quadratic Discriminant Analysis (QDA)

Naive Bayes



LDA for one predictor ($p = 1$)

We assume that $f_k(x)$ is normal:

$$f_k(x) = \frac{1}{\sqrt{2\pi}\sigma_k} \exp\left(-\frac{1}{2\sigma_k^2}(x - \mu_k)^2\right)$$

We also assume that all the classes have the same variance σ

$$Pr(Y = k | X = x) = \frac{\pi_k f_k(x)}{\sum_{l=1}^K \pi_l f_l(x)}$$

$$p_k(x) = \frac{\pi_k \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2\sigma^2}(x - \mu_k)^2\right)}{\sum_{l=1}^K \pi_l \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2\sigma^2}(x - \mu_l)^2\right)}$$



Theoretical Result: Bayes Classifier

The test error is minimized by a very simple classifier that assigns each observation to the most likely class, given its predictor values

That is, we should simply assign the observation with predictor x_0 to the class j for which the following probability is largest:

$$Pr(Y = j|X = x_0)$$

The error made by the Bayes classifier is only the *irreducible* error:

$$1 - E(\max_j Pr(Y = j|X))$$



Using the Bayes Classifier

Under the LDA assumptions (normal density functions with the same variance for all classes), the Bayes classifier assigns an observation x to the class for which the following (called *discriminant*) is the largest:

$$\delta_k(x) = x \cdot \frac{\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + \log(\pi_k)$$

Practically, however, we are not able to calculate the Bayes classifier. We need estimates of μ_k and σ .



Linear Discriminant Analysis (LDA)

To calculate $\delta_k(x) = x \cdot \frac{\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + \log(\pi_k)$, we use the following estimates:

$$\hat{\mu}_k = \frac{1}{n_k} \sum_{i:y_i=k} x_i$$

(average of all training observations in class k)

$$\hat{\sigma}^2 = \frac{1}{n-K} \sum_{k=1}^K \sum_{i:y_i=k} (x_i - \hat{\mu}_k)^2$$

(Weighted average of the sample variances across the K classes)

$$\hat{\pi}_k = \frac{n_k}{n}$$



Linear Discriminant Analysis (LDA)

An observation is assigned to the class for which the following *discriminant* (which is linear in x) is the largest:

$$\hat{\delta}_k(x) = x \cdot \frac{\hat{\mu}_k}{\hat{\sigma}^2} - \frac{\hat{\mu}_k^2}{2\hat{\sigma}^2} + \log(\hat{\pi}_k)$$



LDA: Numerical Example

We want to classify customers as Buyer (class 1) or Non-buyer (class 0) based on a single feature: age.

Training data:

Class 0 (Non-buyers): ages = [20, 25, 30]

Class 1 (Buyers): ages = [40, 45, 50]

So:

$(n_0 = 3), (n_1 = 3), \text{ total } (n = 6)$

Prior probabilities (class proportions):

$$[\pi_0 = \frac{3}{6} = 0.5, \quad \pi_1 = \frac{3}{6} = 0.5]$$



LDA: Numerical Example

Training data:

Class 0 (Non-buyers): ages = [20, 25, 30]

Class 1 (Buyers): ages = [40, 45, 50]

Mean age in each class:

$$\text{Class 0: } [\mu_0 = \frac{20+25+30}{3} = \frac{75}{3} = 25]$$

$$\text{Class 1: } [\mu_1 = \frac{40+45+50}{3} = \frac{135}{3} = 45]$$



LDA: Numerical Example

Training data:

Class 0 (Non-buyers): ages = [20, 25, 30]

Class 1 (Buyers): ages = [40, 45, 50]

For each class, compute the squared deviations from the class mean (will be useful at the next step)

$$\text{Class 0: } [SS_0 = (20 - 25)^2 + (25 - 25)^2 + (30 - 25)^2 = 50]$$

$$\text{Class 1: } [SS_1 = (40 - 45)^2 + (45 - 45)^2 + (50 - 45)^2 = 50]$$

Total within-class sum of squares

$$[50 + 50 = 100]$$



LDA: Numerical Example

In LDA, the pooled within-class variance is typically estimated as:

$$[\hat{\sigma}^2 = \frac{1}{n-K} \sum_{k=1}^K \sum_{i:y_i=k} (x_i - \hat{\mu}_k)^2]$$

Here:

(n = 6) (total observations)

(K = 2) (classes)

$$[\hat{\sigma}^2 = \frac{100}{6-2} = \frac{100}{4} = 25]$$

Common standard deviation:

$$[\hat{\sigma} = \sqrt{25} = 5]$$



LDA: Numerical Example

$$\hat{\delta}_k(x) = x \cdot \frac{\hat{\mu}_k}{\hat{\sigma}^2} - \frac{\hat{\mu}_k^2}{2\hat{\sigma}^2} + \log(\hat{\pi}_k)$$

Classify as 1 if:

$$\delta_1(x) > \delta_0(x)$$

For two classes, this is equivalent to: $x > c$
with:

$$c = \frac{\mu_0 + \mu_1}{2} + \frac{\sigma^2}{\mu_1 - \mu_0} \log\left(\frac{\pi_0}{\pi_1}\right)$$



LDA: Numerical Example

Training data:

Class 0 (Non-buyers): ages = [20, 25, 30]

Class 1 (Buyers): ages = [40, 45, 50]

$$(\mu_0 = 25), (\mu_1 = 45), (\sigma^2 = 25), \pi_0 = \pi_1 = 0.5 \Rightarrow \log(\pi_0/\pi_1) = \log(1) = 0$$

$$\text{So } c = \frac{25+45}{2} = \frac{70}{2} = 35$$

LDA Classification rule:

If age $x > 35 \rightarrow$ predict *Buyer (class 1)*

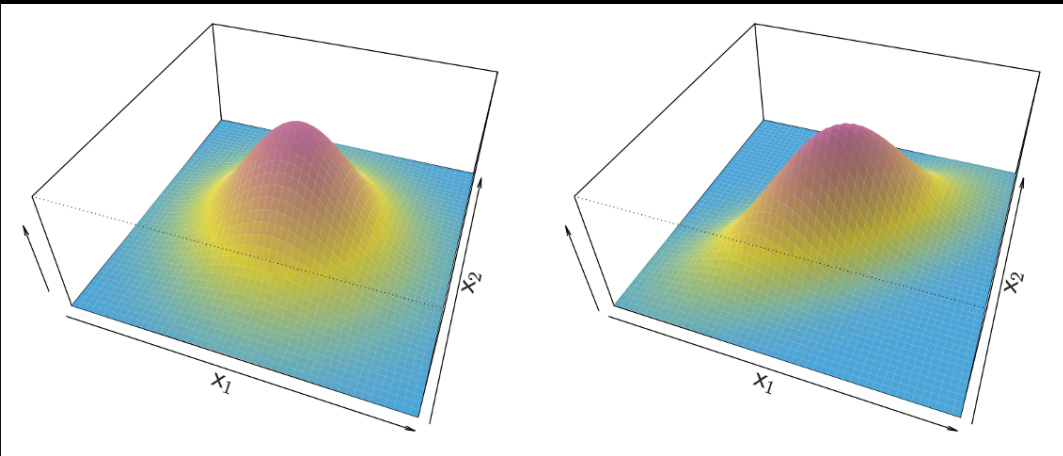
If age $x \leq 35 \rightarrow$ predict *Non-buyer (class 0)*



LDA for $p > 1$

Assume $X = X_1, \dots, X_p$ is drawn from a multivariate Gaussian with class-specific mean vector and common covariance matrix Σ :

$$f(x) = \frac{1}{(2\pi)^{p/2} |\Sigma|^{1/2}} \exp \left(-\frac{1}{2} (x - \mu)^T \Sigma^{-1} (x - \mu) \right)$$





LDA for $p > 1$: Using the Bayes Classifier

The Bayes classifier assigns an observation $x = (x_1, \dots, x_p)$ to the class k for which the following discriminant is the largest:

$$\delta_k(x) = x^T \Sigma^{-1} \mu_k - \frac{1}{2} \mu_k^T \Sigma^{-1} \mu_k + \log \mu_k$$

We need to estimate the unknown parameters $\mu_1, \dots, \mu_k, |\Sigma, \pi_1, \dots, \pi_k$

We can generalize the same formulas for $p = 1$



What is Σ ?

Multivariate normal

$$p(\mathbf{x}; \mu, \Sigma) = \frac{1}{(2\pi)^{\frac{n}{2}} |\Sigma|^{\frac{1}{2}}} \exp \left(-\frac{1}{2} (\mathbf{x} - \mu)^T \Sigma^{-1} (\mathbf{x} - \mu) \right)$$

Multivariate normal (p=2)

$$X \sim \mathcal{N}(\mu_X, \sigma_X^2) \quad Y \sim \mathcal{N}(\mu_Y, \sigma_Y^2)$$

$$\rho(X, Y) = \rho = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \text{Var}(Y)}} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} \Rightarrow \text{Cov}(X, Y) = \rho \sigma_X \sigma_Y$$

$$\boldsymbol{\mu} = \begin{bmatrix} \mu_X \\ \mu_Y \end{bmatrix} \quad \Sigma = \begin{bmatrix} \sigma_X^2 & \rho \sigma_X \sigma_Y \\ \rho \sigma_X \sigma_Y & \sigma_Y^2 \end{bmatrix}$$

$$f_{X,Y}(x, y) = \frac{1}{2\pi \sigma_X \sigma_Y \sqrt{1 - \rho^2}} \exp \left(-\frac{z}{2(1 - \rho^2)} \right), \quad x, y \in \mathbb{R}$$

$$z = \frac{(x - \mu_X)^2}{\sigma_X^2} - \frac{2\rho(x - \mu_X)(y - \mu_Y)}{\sigma_X \sigma_Y} + \frac{(y - \mu_Y)^2}{\sigma_Y^2}$$



Common Variance Assumption in LDA

In LDA, σ and Σ are not a function of the class k

The variance is *common* across all classes

$$\hat{\delta}_k(x) = x \cdot \frac{\hat{\mu}_k}{\hat{\sigma}^2} - \frac{\hat{\mu}_k^2}{2\hat{\sigma}^2} + \log(\hat{\pi}_k)$$

$$\hat{\delta}_k(x) = x^T \Sigma^{-1} \mu_k - \frac{1}{2} \mu_k^T \Sigma^{-1} \mu_k + \log \mu_k$$



Next Time

QDA

Naive Bayes

(Poisson Regression)



Evaluating the Performance of a (Binary) Classifier

Confusion Matrix

Accuracy

Precision

Recall

F1 Score

AUC



Confusion Matrix (Binary Classification)

	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

- TP: correctly predicted positives
- TN: correctly predicted negatives
- FP: negatives incorrectly predicted as positive
- FN: positives incorrectly predicted as negative



Accuracy

Definition: Fraction of all predictions that are correct.

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}$$

Interpretation:

“Out of all instances, how often is the model right?”

Simple and intuitive, but can be misleading with imbalanced classes.



Precision

Definition: Fraction of predicted positives that are truly positive.

$$\text{Precision} = \frac{TP}{TP+FP}$$

Interpretation:

“When the model says ‘positive’, how often is it correct?”

Important when **false positives are costly** (e.g., spam detection, fraud alerts).



Recall (Sensitivity, TPR)

Definition: Fraction of actual positives that are correctly identified.

$$\text{Recall} = \frac{TP}{TP+FN}$$

Interpretation:

“Of all real positives, how many did the model catch?”

Important when **false negatives are costly** (e.g., disease screening, fault detection).



F1 Score

Definition: Harmonic mean of precision and recall.

$$F_1 = 2 \cdot \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2 TP}{2 TP + FP + FN}$$

Interpretation:

Balances precision and recall in a single number.

Useful when classes are **imbalanced** and you care about both FP and FN.



Visual Summary

Metric	Formula	Focus
Accuracy	$\frac{TP+TN}{TP+TN+FP+FN}$	Overall correctness
Precision	$\frac{TP}{TP+FP}$	Quality of positive predictions
Recall	$\frac{TP}{TP+FN}$	Coverage of actual positives
F1 Score	$2 \cdot \frac{P \times R}{P+R}$	Balance between P and R



When to Use Which Metric?

Accuracy

Classes are roughly balanced

All types of errors have similar cost

Precision

False positives are expensive or harmful

You want high confidence in positive predictions



When to Use Which Metric?

Recall

False negatives are expensive or harmful

You want to miss as few positives as possible

F1 Score

Imbalanced classes

You need a single metric reflecting both precision and recall



Evaluating the Performance of a Classifier

Confusion Matrix of LDA predictions on the Default dataset
(10,000 observations)

		<i>True default status</i>		
		No	Yes	Total
<i>Predicted default status</i>	No	9644	252	9896
	Yes	23	81	104
	Total	9667	333	10000



Problems with Intuition Behind Generative Models

These methods approximate the Bayes Classifier, which aims at minimizing the total classification error (= on all classes!)

For a bank, misclassifying a person who will default is often more critical than misclassifying a person who will not default

		<i>True default status</i>		
		No	Yes	Total
<i>Predicted default status</i>	No	9644	252	9896
	Yes	23	81	104
	Total	9667	333	10000



How Does the Bayes Classifier Work?

The Bayes classifier assigns an observation to the class for which the posterior probability $p_k(X)$ (or, better, its estimate) is the largest.

For $K = 2$ (binary classification) and $p = 1$, this means to use the threshold 50% on the posterior probability

$$Pr(\text{default} = \text{Yes} | X = x) > 0.5$$



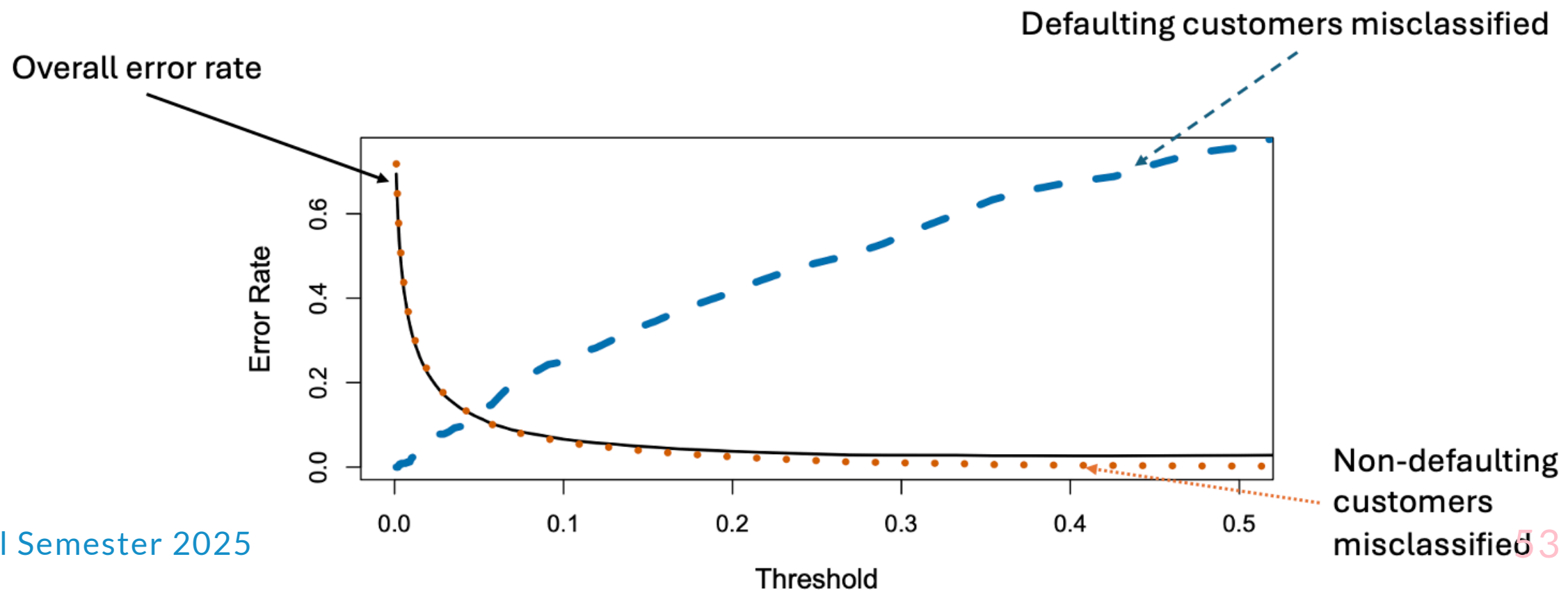
Can we Adjust the Classification Threshold?

Yes, but what does it mean to do it?

$$Pr(\text{default} = \text{Yes} | X = x) > 0.2$$

0.5				
		<i>True default status</i>		
		No	Yes	Total
<i>Predicted default status</i>	No	9644	252	9896
	Yes	23	81	104
	Total	9667	333	10000

0.2				
		<i>True default status</i>		
		No	Yes	Total
<i>Predicted default status</i>	No	9432	138	9570
	Yes	235	195	430
	Total	9667	333	10000





Terminology!

		<i>True class</i>		
		– or Null	+ or Non-null	Total
<i>Predicted class</i>	– or Null	True Neg. (TN)	False Neg. (FN)	N*
	+ or Non-null	False Pos. (FP)	True Pos. (TP)	P*
	Total	N	P	

Name	Definition	Synonyms
False Pos. rate	FP/N	Type I error, 1–Specificity
True Pos. rate	TP/P	1–Type II error, power, sensitivity, recall
Pos. Pred. value	TP/P*	Precision, 1–false discovery proportion
Neg. Pred. value	TN/N*	

Default example: + (positive) is the “defaulting” class (customers who cannot repay their CC debt)

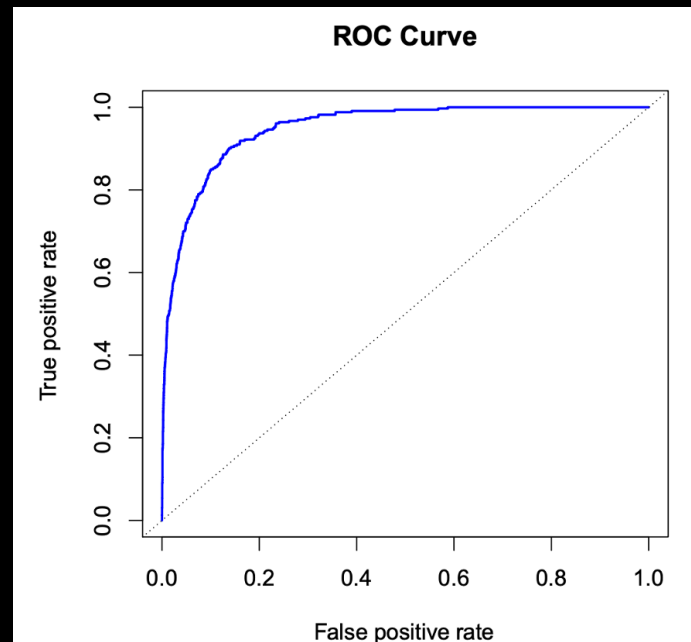


The ROC Curve

ROC: Receiver Operating Characteristics

Plots two types of performance measures as a function of the classification threshold

(False positive: non-defaulters classified as defaulters)





When Is a Classifier a Good One?

When its Area Under the ROC Curve (AUC) is close to 1

